GÖRKEM MERİÇ

DYNAMIC ANALYSIS ENGINEER

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Ankara/TURKEY



ABOUT

Hardworking Engineer graduated from Hacettepe University, Mechanical Engineering Department, Automotive Program. Strong abilities in critical thinking, teamwork, problem-solving, and communication. Currently working in the defense industry on helicopter rotors. In the long term, it is aimed to improving theoretical background by pursuing master's and doctorate degrees. Main personal interests include fitness, basketball, reading, and board games.

INDUSTRIAL EXPERIENCE

EXPERT ROTOR DYNAMICS ANALYSIS ENGINEER

10.2022 - Present

TURKISH AEROSPACE
ANKARA | ON SITE

This role mainly focuses on the analysis of dynamics of the helicopter rotor and qualification of subsytem equipments in terms of vibration. The key responsibilities are as follows:

- Performing multibody dynamics analysis in Dymore in terms of load distribution, stability characteristics, modal characteristics, and kinematics, and reporting outcomes.
- Preparing certification documents with respect to CS-29 and presenting them to the authorities.
- Planning and attending flight tests, monitoring real-time data for flight safety, development, and certification at all project stages.
- Preparing component spectrum in terms of load or motion for technical specification by combining analysis and flight test results.
- Post-processing flight data to correlate with mathematical models.
- Developing GUI's for visualization of the post-processed data.
- Developing tools for data interpretation and supporting data processing using MATLAB and Python.
- Participating in environmental qualification processes for subsystem equipment in terms of vibration by preparing vibration test profiles according to DO-160 or MIL-STD-810 and attending vibration tests.
- Examining vibration levels from flight tests and comparing them with established vibration qualification tests.

MANUFACTURING INTERN ENGINEER

08.2021 - 09.2021

NUROL MAKINA
ANKARA | ON SITE

- Examined received work packages and technical drawings for assigned processes alongside technicians.
- Spent three days in each of the manufacturing stages in the factory, observing all techniques: press brake, CNC machining, laser cutting, and automated welding.
- Identified missing vehicle parts in the assembly line.

ANALYSIS, SIMULATION & TEST INTERN ENGINEER

07.2021 - 08.2021

NUROL MAKINA
ANKARA | ON SITE

- Observed CAE processes using ANSYS and analysis of complex, non-uniform structures.
- Attended several field tests on prototype vehicles involving various mechanical/electrical components.

EDUCATIONAL BACKGROUND

MASTER OF SCIENCE IN MECHANICAL ENGINEERING 02.2024 - Present

HACETTEPE UNIVERSITY
ANKARA

BACHELOR'S DEGREE IN AUTOMOTIVE ENGINEERING 09.2016 - 05.2022

HACETTEPE UNIVERSITY
ANKARA | 3.14/4.00

PROJECT PARTICIPATIONS

MINIATURE SUBMARINE FINAL DESIGN PROJECT 09.2021 - 05.2022

Associated with Hacettepe University

- System Design:
  - Participated in designing Ballast System (Static Diving), Maneuvering System (Dynamic Diving), Thrust System, Internal Electronic System and Wired Controller Unit
- Programming:
  - Participated in designing the algorithms of microcontroller to get all subsystems work together and properly under given inputs.
- Manufacturing and Testing:
  - Participated in manufacturing and testing process of submarine. Main objective was to assemble 3D printed parts and all electronics into submarine and testing impermeability of assemblies.

TUBITAK INTERNATIONAL UNMANNED AERIAL VEHICLES | ROTARY WINGS 01 - 09.2019

Associated with TEKNOFEST

- System Design:
  - Participated in design process of the compact frame of the UAV by considering given dimension/weight ratio. Also, participated in desing process of a rack-and-pinion system for a specific task which taking a object from the ground level by an electromagnet.
- Programming:
  - Participated in designing the Arduino algorithms to move the rack-and-pinion system under given conditions by another microcontroller.
- Manufacturing and Testing:
  - Participated in manufacturing and assembling procedure of the frame parts. Also, took a role of testing the several UAV prototypes for competition tasks.

REFERENCES

EMIR KUTLUAY - THESIS SUPERVISOR

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HACETTEPE UNIVERSITY MECHANICAL ENGINEERING DEPARTMENT
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SELCUK HIMMETOGLU - ACADEMIC

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ARIF BATUHAN YILMAZ - COWORKER

EXPERT DESIGN ENGINEER
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